

State Machine Question (100 marks)

Given the problem description below, specify the **state machine** for the software of a self-service checkout. **Show the states as well as transitions including events, guards, and actions. Define composite states when needed to reduce complexity but do not use orthogonal regions.** You do **not** need to specify parameters for events, guards, and actions. Use **meaningful** method names for guards and actions in your state machine. Method names must clearly show what the system needs to do or to which value a variable is set. You do **not** need to define the method bodies of those methods.

Problem Description: Self-Service Checkout SSC7

The Self-Service Checkout SSC7 is used by supermarkets to allow customers to scan their purchases and pay for them, often without any help of supermarket staff.

As shown in the figure on the right, the SSC7 consists of the following parts: (i) an area to the left to place items that still need to be scanned, (ii) a barcode reader, (iii) a touch screen, (iv) a scale in front of the touch screen to weigh items, (v) an area to the right with an integrated security scale where items are placed that have been scanned, (vi) a payment terminal to pay by credit card, (vii) a printer for the bill, and (viii) a light to indicate to a staff member that they are needed.



The SSC7 Software (SSC7S) interacts with parts (ii) to (viii). The barcode reader provides a scanned number to the SSC7S. The SSC7S uses the touch screen to show all information to the customer, and the customer presses buttons and provides the four-digit product code of items without barcode to the SSC7S. Both scales, the one in the front of the touch screen and the security scale, provide a weight to the SSC7S. The SSC7S sends a payment request to the payment terminal, which sends back a success message with an authorization code, a fail message with an error because the payment failed, or a cancel message because the customer canceled the transaction instead of using their credit card. Furthermore, the SSC7S sends the bill to the printer and turns the light for the staff member on and off.

A customer can either scan the barcode of an item or enter the four-digit product code for items without barcode. If the barcode or the product code do not exist in the system, an error message is shown to the customer. When a correct barcode is scanned, then the item is shown on the touch screen. When a correct product code is entered, the item needs to be weighed and afterwards it is shown with its weight on the touch screen. In both cases (i.e., item with or without barcode), the item must then be placed in the area on the right side for a security weight check. If the weight matches the expected weight, then the system is ready for the next item. If it does not, staff must be called for an override to clear the item from the bill. When it is possible to scan an item and the bill has at least one item, the customer may indicate that they want to remove an item from the bill, which also requires staff to be called for an override to clear the item from the bill.

A customer may also indicate that they brought their own bag(s). Each bag has to be weighed with the security scale. This can be done when it is also possible to scan an item. After a bag is weighed, the number of bags is indicated on the touch screen and the system is ready for the next item.

The customer starts the payment process by pressing the Pay button on the touch screen. If the bill has at least one item, the system sends the payment request to the payment terminal and waits for the payment result. If it is successful, the system prints the bill and readies the system for the next customer. If it is unsuccessful, the system is ready for the next item (and shows an error message if the payment failed).

The customer may cancel the weighing of an item or bag, in which case the system is ready for the next item. When it is possible to scan an item or when the customer is required to place the item (not a bag) in the area on the right side for a security weight check, the customer may also request to cancel everything and start over, which requires staff to be called for an override to clear everything.

When an override is required, the system turns on the light and waits for the staff to enter their override code. If the code is incorrect, an error message is shown. Otherwise, the light is turned off, and either the item or everything is cleared.

When the system is idle for a certain time, the SSC7S receives a timeout warning signal. This only happens when staff has not been called for an override and the payment process has not yet been started. The timeout warning signal starts a 60 second countdown on the touch screen. If the customer presses the Continue button, they continue where they had left off. If the countdown reaches 0, the system clears everything and readies the system for the next customer.